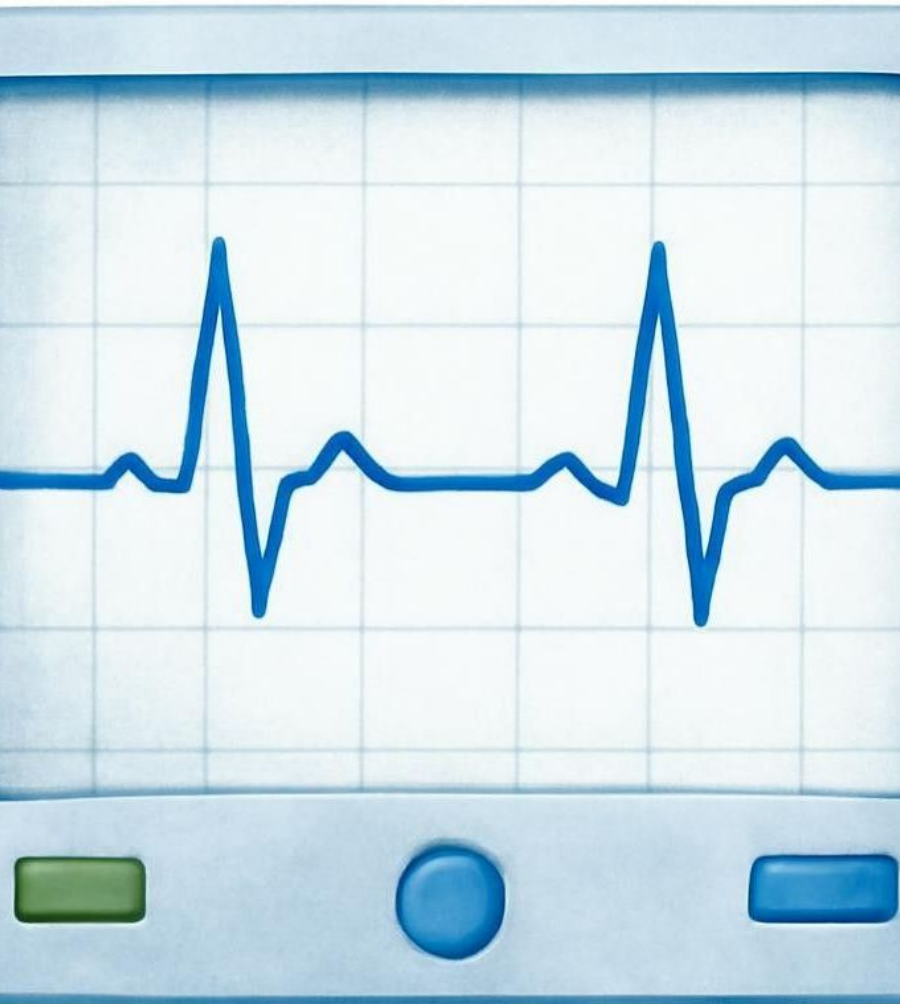




Heart Disease Prediction using Machine Learning

Capstone Project | Machine Learning & AI Virtual Internship

End-to-end binary classification pipeline designed to predict the likelihood of heart disease in patients based on clinical diagnostic features.

Dataset & Data Preprocessing

Dataset

UCI Heart Disease Dataset (Kaggle) — a well-established benchmark for cardiovascular risk classification.

Data Cleaning

Evaluated feature integrity, verified null values, and cleaned incomplete entries to ensure a reliable training foundation.

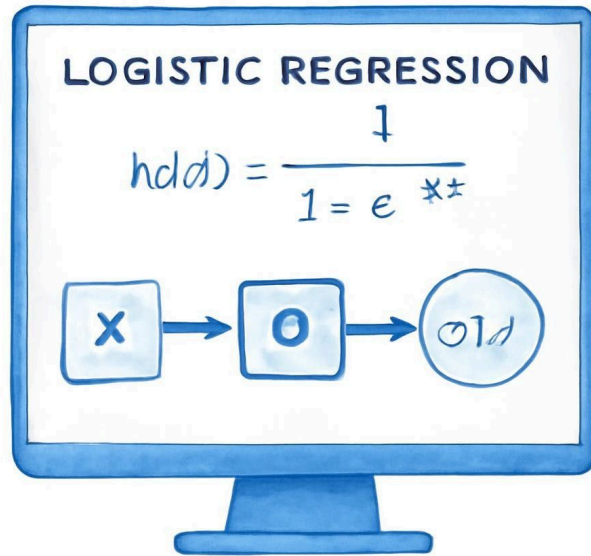
Feature Encoding

Converted categorical health attributes (e.g., chest pain type, resting ECG) into numerical format using One-Hot Encoding (`pd.get_dummies`).

Feature Scaling

Applied `StandardScaler` to normalize numerical variance across biometric features (e.g., cholesterol, blood pressure, max heart rate).





Model Architecture & Training

1 Algorithm

Logistic Regression (Scikit-Learn)
— a proven, interpretable
classifier well-suited for binary
clinical outcomes.

2 Data Splitting

80% Training Set / 20% Testing Set with a fixed random state for full reproducibility across experiments.

3 Model Rationale

Selected for its high interpretability in clinical diagnosis, outputting well-calibrated probabilistic scores that clinicians can act on.

Model Evaluation & Performance Metrics

85.25%

Accuracy

High overall predictive correctness across all patient cases.

87.10%

Precision

High confidence when diagnosing positive heart disease cases, minimizing false alarms.

84.38%

Recall / Sensitivity

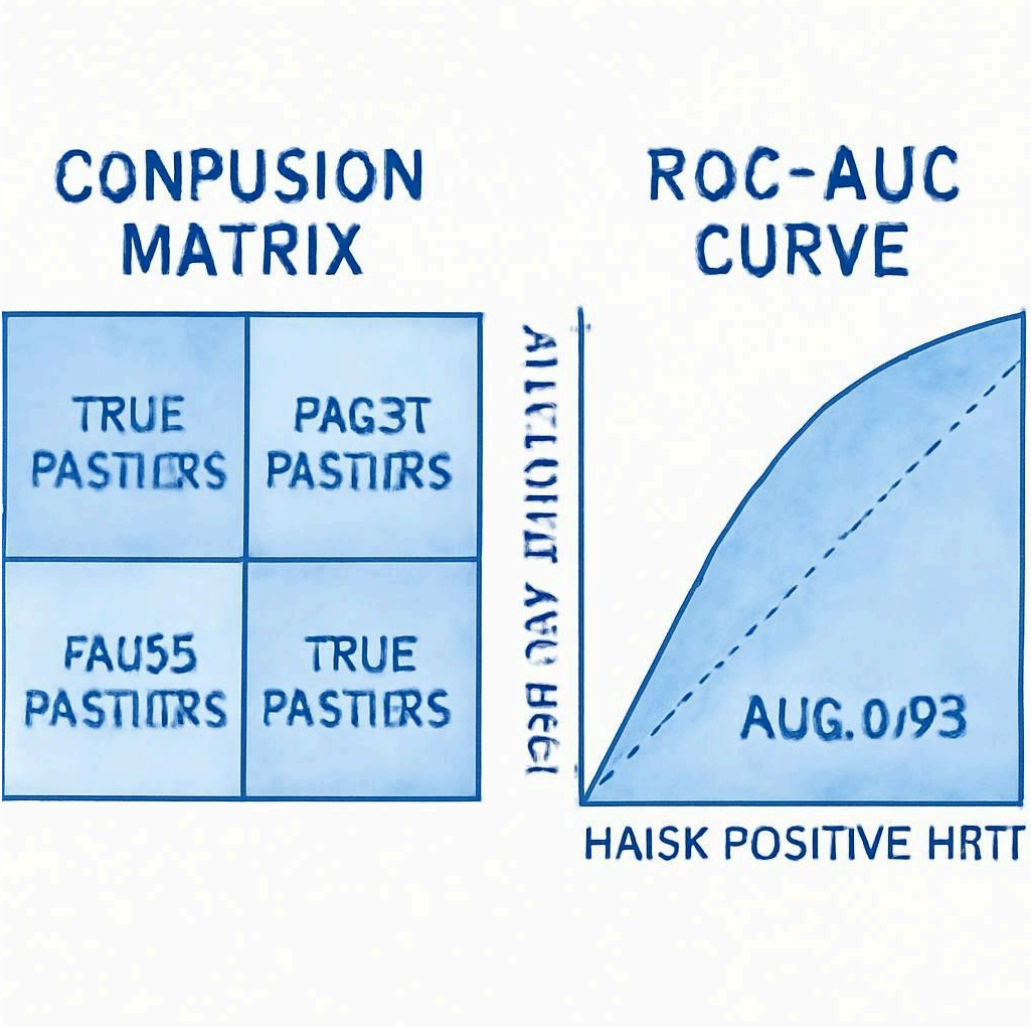
Effective identification of high-risk patients to minimize missed diagnoses.

0.93

ROC AUC Score

Demonstrates exceptional class-separation capability across all thresholds.

Visual Diagnostics & Clinical Insights



Confusion Matrix Analysis

Visualizes the distribution of **true positives**, **true negatives**, **false positives**, and **false negatives** — providing a granular view of where the model succeeds and where clinical caution is warranted.

ROC-AUC Curve

Highlights a high true-positive rate across varying classification thresholds, confirming the model's robustness at **AUC = 0.93** — well above the 0.5 random baseline.

Impact & Conclusion

This pipeline serves as a **reliable clinical decision-support tool**, assisting doctors in early cardiovascular risk detection and enabling timely, data-driven intervention for at-risk patients.

- ✔ An AUC of 0.93 positions this model as a strong candidate for real-world clinical screening support.